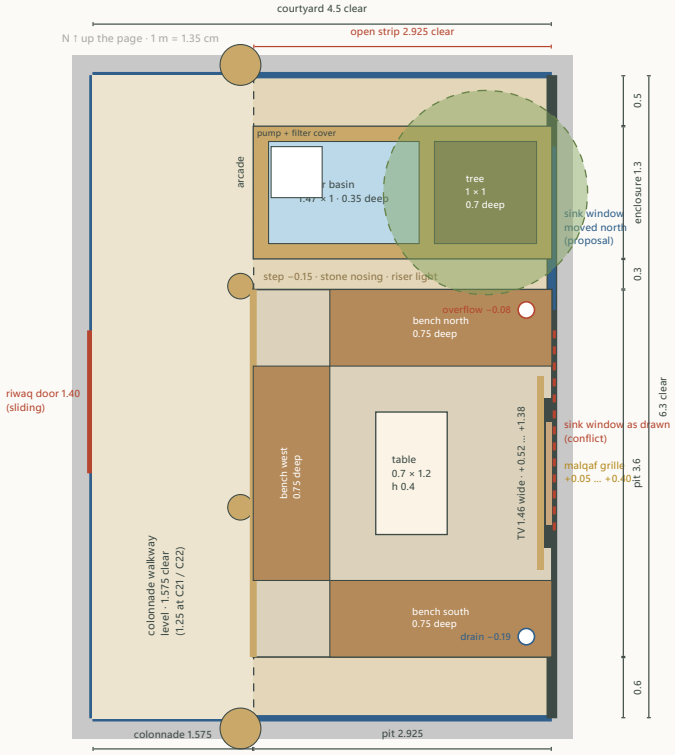
Courtyard layout · the room open to the sky · Taif / Palm House V7.1

Owner decisions 1 Nov 2026 (design review Y-02 / Y-03 / Y-04 / E-03) · clear dimensions between finished faces · the data in `3d/character.yaml` `courtyard_layout` and `studies/air_system.typ` (the grille) · nothing here is for construction



1 · Circulation and the real clear widths

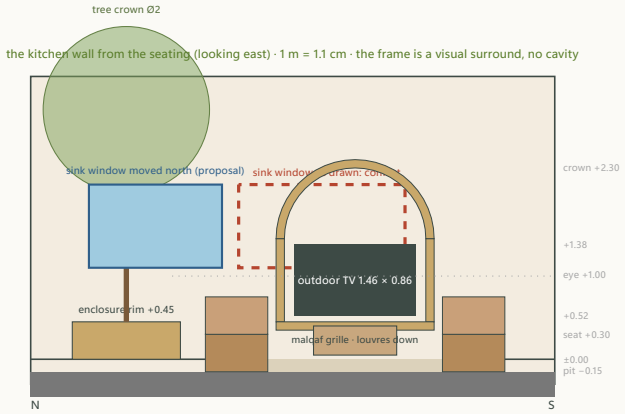
- **Courtyard:** 4.5 × 6.3 m between the finished faces (28.35 m²): the riwaq glass face x 4.35, the kitchen wall face x 8.85, the women's majlis glass y 6.6, the living glass y 12.9.
- **Colonnade walkway** under the projection, level, kept clear: 1.575 m from the riwaq glass to the soffit edge (x 5.925); 1.325 m at the two arcade columns (Ø0.25), 1.25 m at C21 / C22 (sleeves Ø0.4). The riwaq door (sliding, 1.40) opens onto it; it links the living's glass, the women's majlis glass and the riwaq.
- **Open strip east of the colonnade: 2.925 m** clear from the soffit edge to the kitchen wall (not 3.15 on axes; 2.85 m at the C21 / C22 sleeves).

2 · Seating one step down

- **The pit:** from the colonnade edge to the kitchen wall, 2.925 × 3.6 m (10.53 m²), floor −0.15, **one step** of 0.15 (the owner's choice); no walkway round it: the north and south strips are the cleaning strips at the glass (0.6 m at the living, 0.5 m at the women's majlis behind the enclosure) and the 0.3 m between the enclosure and the step.
- **Benches:** built-in, 0.75 deep (seat 0.45 above the pit floor = +0.30, back to +0.75), stone bases with cushions: west bench 2.1 m on the step edge facing the TV, with 0.75 m gaps at both ends to step down; north and south benches 2.17 m from the kitchen wall. Seat length 6.45 m: **4 comfortable with room to spare (7 places at 0.9 m each, 10 at 0.6 m)**; the east side stays free for the TV, the grille and cleaning (1.02 m).
- **Table:** 0.7 × 1.2 m, 0.4 high (meals), leg room 0.45 m west, 0.45 / 0.45 m north / south.
- **Step safety:** a contrasting honey-stone nosing 8 cm wide along the whole step edge, a continuous linear light (IP67 LED, warm) in the riser, no loose rugs on the edge; the bench backs (+0.75) edge the pit on three sides.

Study checks passed (9)

3 · The TV wall (the kitchen wall, seen from the seating) · 4 · The north enclosure: water and tree



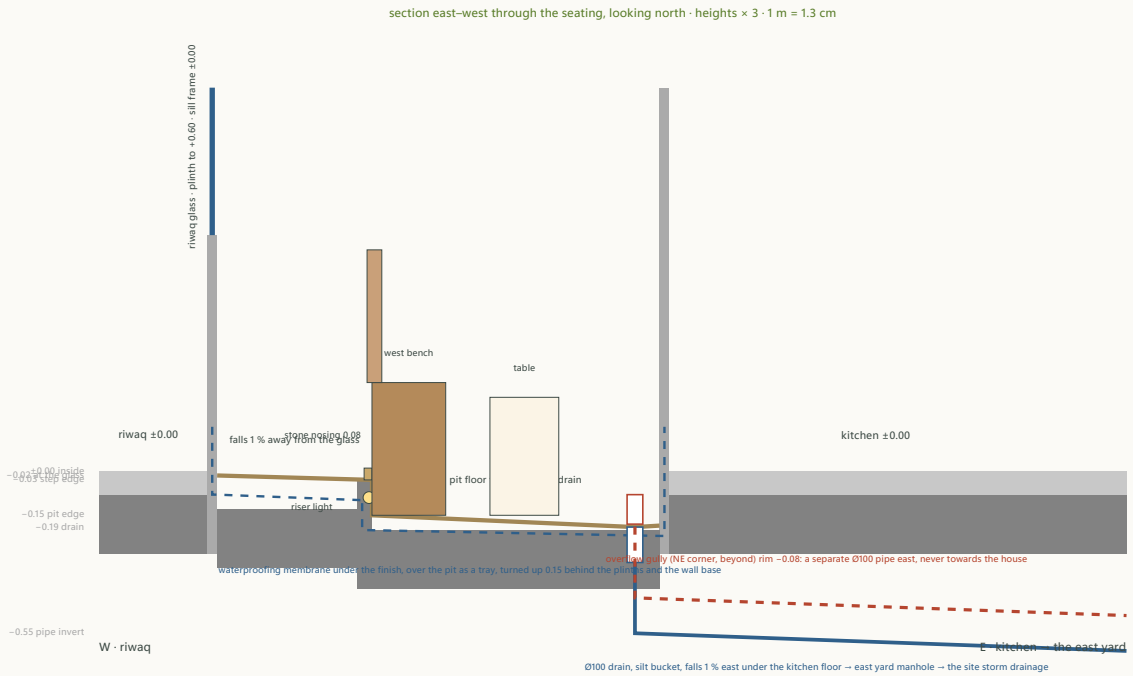
3 · The TV, its frame and the grille

- **The TV:** an **outdoor-rated** set (sun, rain and dust: an IP55 / IP56 “partial-sun” outdoor model, 65” class, ≈ 1.46 × 0.86 m; the final size, bracket, ventilation and installation clearances **per the chosen model**), centred on the seating at y 10.5, bottom +0.52 (0.67 above the pit floor), centre +0.95 against a seated eye at +1.00: the screen sits a little below eye level at 2.17 m from the west bench. Power and signal in a sleeved conduit from the kitchen; the fountain switch beside it.
- **The frame:** honey stone, a visual surround only: two jambs and a semicircular head, band 0.1 wide, projecting 0.06 from the wall, inner width 1.7 (0.12 each side of the TV), springing +1.45, crown +2.30 no tight cavity: the TV hangs on its bracket 0.06 m proud of the wall with the clearances the model asks (air behind and above it).
- **The grille** (`studies/air_system.typ`): below the TV, y 10–11, +0.05 … +0.40 (0.2 above the pit floor), low and decorative, removable for cleaning, its louvres turned down so the night air runs along the floor and not into the faces on the benches; 1.6 m from the fountain enclosure; no bench or cushion in front of it (the east side of the pit stays free). The chamber under the kitchen floor moves with it to y 9.5–10.5, the duct 8.5 m long.
- **The conflict:** the **kitchen sink window** as drawn (y 9.1–11.1, +1.10 … +2.10) sits exactly where the TV and its frame go: the TV overlaps it, the arched frame covers it. **Proposal (owner decision, not applied):** move the sink window north to y 7.3–8.9 (1.6 m wide, same sill +1.10), over the north part of the sink counter (which runs y 7.5–12.9): the sink looks at the fountain and the tree, the solid wall y 8.9–12.9 takes the TV and its frame. The alternative, a smaller TV below the sill without the frame, loses the arch.

4 · The north enclosure: one stone box, two basins

- **The box:** honey stone, x 5.93–8.85, y 7.1–8.4 (2.93 × 1.3 outside, walls 0.15, rim at +0.45 = a seat-height ledge), 0.5 m off the women’s majlis glass (cleaning), 0.3 m from the step.
- **Water basin** (west part): 1.47 × 1 inside, 0.35 deep, water at +0.25 a **laminar lip** in honey stone on the divider wall at +0.60, the water sheeting over stone into the basin (no spray, no mist); its own drain (bottom) and an overflow standpipe at the water level; the pump and the filter in a dry chamber under the access cover (0.5 × 0.5) at the west end; a **switch by the TV** turns the fountain off for watching; a GFCI / RCD circuit.
- **Planting basin** (east part, by the kitchen wall): 1 × 1 inside, 0.7 m of soil over a drainage layer and its own drain (adjust the depth to the chosen specimen); the basin tanked and separated from the water basin by the 0.15 divider; drip irrigation on a timer.
- **The tree:** medium, light canopy, bought at a good size, **fixed**: **Laurus nobilis** trained as a standard, or a clipped **Ficus nitida**; the final choice by a Taif nursery (climate, the winter cold, the irrigation water). Assumed: clear stem 2 m, crown Ø2 centred 3 m up, kept clipped off the kitchen wall and the glass.
- **Light check (simple):** the crown stands 1.15 m from the women’s majlis glass and covers 0.9 m of its 2.6 m width above +2.00: ≈ 11 % of that glass area in shade, the rest and the glass below +2.00 clear; the living’s glass is 5.15 m away: no shade. **To verify on the V7 3D model** (the sun paths) once the tree’s size is chosen.

6 · Drainage: the courtyard floor, the seating floor, the thresholds, the overflow and the discharge · section east–west through the seating (y ≈ 10.5)



The drainage, point by point

- **Thresholds:** the inside floors at ±0.00; the courtyard finish at the three glass lines and the riwaq door at **−0.02** (a 15 mm step, within the 15 mm of R-08); the glass sills on a plinth with the membrane turned up 0.15 behind it; a slot drain is not needed: the floor falls away from every door.
- **Falls:** the colonnade and the north / south strips fall 1 % away from the glass to the step edge (−0.03) and to the pit; the pit floor falls 1 % from its west edge (−0.15) to the **floor drain** at the south-east corner (−0.19, x 8.6, y 12.1): a Ø100 gully with a removable grating and a silt bucket, cleanable from the pit.
- **Overflow, independent:** a second gully at the north-east corner (x 8.6, y 8.9) with its rim at **−0.08**, 11 cm above the drain and below the step edge (−0.03): if the drain blocks, the pit fills to the overflow and the water leaves by a separate pipe east; it can never reach the glass lines, which stand 6.5 cm higher.
- **Pipes:** both run east under the kitchen floor (invert −0.55 at the drain, 1 % fall) across the kitchen (north of the dining door, south of the malqaf chamber) to the east wall and the **east yard manhole**, then to the **site storm drainage** (the plumbing strategy: everything east drains to the east yard; the plumbing study, open item 9, sets the manhole and the outfall). The kitchen's sealed malqaf cover is at y 9.7–10.3: the pipes pass north and south of the chamber, not through it.
- **The fountain:** the water basin's bottom drain (−0.35) and its overflow standpipe (at the water level +0.25) join the overflow pipe; the planter's drain (bottom −0.70) joins it too, through a gravel layer and a geotextile; the pump chamber has its own small drain. The water basin is filled from a tap in the pump chamber; the fountain runs on a timer and the switch by the TV.
- **Waterproofing:** one membrane over the whole courtyard slab (lowered 0.05 for the outdoor build-up), over the pit as a tray with the step as an upstand, up 0.15 behind the glass plinths and the kitchen wall base; the enclosure's two basins tanked separately; the finish honey-light stone (slip-resistant outdoors), the step nosing honey stone.
- **Engineer / MEP to confirm:** the slab recess for the pit (0.15) and the planter (0.75), the gully and pipe sizes, the east yard manhole and the outfall.

Dimensions (clear, between finished faces)

Item	m	Note
Courtyard clear	4.5 × 6.3	28.35 m²; axes 4.70 × 6.50
Colonnade walkway (level)	1.575 / 1.325 / 1.25	to the soffit edge / at the arcade columns / at C21–C22
Open strip east of the colonnade	**2.925**	the soffit edge x 5.925 → the kitchen wall face x 8.85 2.85 at the sleeves
Seating pit	2.925 × 3.6	10.53 m², floor −0.15, one step 0.15
Benches	0.75 deep, 6.45 long	seat +0.30, back +0.75 7 comfortable, 10 max
Table	0.7 × 1.2 × 0.4	leg room 0.45 W, 0.45 N, 0.45 S; 1.02 free to the wall
Cleaning strips	0.6 S · 0.5 N	at the living glass / behind the enclosure at the women's majlis glass; 0.3 enclosure → step
Step	0.15	nosing 0.08 honey stone, riser light, no rugs
TV	1.46 × 0.86	y 9.77–11.23, +0.52 … +1.38 per the chosen model
TV frame	1.7 inside, band 0.1	springing +1.45, crown +2.30, proud 0.06
Grille	1 × 0.35	y 10–11, +0.05 … +0.40 0.12 below the TV

Enclosure	2.93 × 1.3	walls 0.15, rim +0.45
Water basin	1.47 × 1 × 0.35	water +0.25, lip +0.60
Planting basin	1 × 1 × 0.7	tree crown Ø2 at 3 m
Levels	−0.02 / −0.03 / −0.15 / −0.19	glass line / step edge / pit edge / drain; overflow rim −0.08 pipe invert −0.55

Conflicts and decisions

- **The kitchen sink window** (page 2): it sits where the TV goes. **Owner decision:** move it north to y 7.3–8.9 (recommended) or take a smaller TV below the sill without the frame. Not applied.
- **The 3D character data** now carry the new layout (`courtyard_layout`); the builder (`character_3d.py`) still draws the old sunken seating, fountain and pergola until it is updated at the next 3D build.
- **The pergola** is removed from the design (owner): the courtyard enclosure study's note and the character data; the roof keeps its guards.
- **Children's play** moves to the garden (R-23 / R-30 updated); the courtyard is for sitting, meals and the TV.